

DSC540WeekProjectMilestone4

May 22, 2025

1 Project Milestone 4

```
[148]: pip install pyspark
```

```
Collecting pyspark
  Downloading pyspark-4.0.0.tar.gz (434.1 MB)
    434.1/434.1
MB 3.0 MB/s eta 0:00:0000:0100:05
  Preparing metadata (setup.py) ... done
Collecting py4j==0.10.9.9 (from pyspark)
  Downloading py4j-0.10.9.9-py2.py3-none-any.whl.metadata (1.3 kB)
  Downloading py4j-0.10.9.9-py2.py3-none-any.whl (203 kB)
    203.0/203.0 kB
2.5 MB/s eta 0:00:00a 0:00:01
Building wheels for collected packages: pyspark
  Building wheel for pyspark (setup.py) ... done
  Created wheel for pyspark: filename=pyspark-4.0.0-py2.py3-none-any.whl
  size=434741237
  sha256=914e45d49bfa129240bb857adf5a868292ccec075db8c8a0c5830d0be01b4c48
  Stored in directory: /Users/vijsharm/Library/Caches/pip/wheels/2d/77/9b/12660b
  e70f7f447940a0caede37ae208b2e0d1c8487dce52a6
Successfully built pyspark
Installing collected packages: py4j, pyspark
Successfully installed py4j-0.10.9.9 pyspark-4.0.0
Note: you may need to restart the kernel to use updated packages.
```

```
[77]: # Importing Pandas packages required for this exercise.
```

```
import pandas as pd
from urllib.request import urlopen
import json

import urllib.request, urllib.parse, urllib.error
import requests
from bs4 import BeautifulSoup
import ssl
import re
```

```
import matplotlib.dates as mdates
from matplotlib.dates import DateFormatter
import matplotlib.ticker as tick
```

```
[79]: def read_json_from_web_api(url):
        """
        Reads a JSON file from a web API.

        Args:
            url (str): The URL of the web API endpoint.

        Returns:
            dict: The JSON data as a Python dictionary, or None if an error occurs.
        """
        try:
            response = requests.get(url)
            response.raise_for_status() # Raise HTTPError for bad responses (4xx
            ↪or 5xx)
            return response.json()
        except requests.exceptions.RequestException as e:
            print(f"Request error: {e}")
            return None
        except json.JSONDecodeError as e:
            print(f"JSON decode error: {e}")
            return None
```

```
[107]: # call Covid 19 API and load it in Json Format
url = "https://api.covidtracking.com/v1/us/daily.json" # Replace with the
            ↪actual API endpoint
data = read_json_from_web_api(url)
df = pd.json_normalize(data)
df = df.drop('hash',axis=1)
df.head(10)
```

```
[107]:
```

	date	states	positive	negative	pending	hospitalizedCurrently	\
0	20210307	56	28756489.0	74582825.0	11808.0	40199.0	
1	20210306	56	28714654.0	74450990.0	11783.0	41401.0	
2	20210305	56	28654639.0	74307155.0	12213.0	42541.0	
3	20210304	56	28585852.0	74035238.0	12405.0	44172.0	
4	20210303	56	28520365.0	73857281.0	11778.0	45462.0	
..	...	...	...	...	...	...	
95	20201202	56	13925720.0	50105551.0	14368.0	100327.0	
96	20201201	56	13722291.0	50764325.0	8764.0	98814.0	
97	20201130	56	13541108.0	50500626.0	14883.0	96134.0	
98	20201129	56	13391077.0	50156395.0	14860.0	93357.0	
99	20201128	56	13253823.0	49929607.0	14780.0	91762.0	

	hospitalizedCumulative	inIcuCurrently	inIcuCumulative \
0	776361.0	8134.0	45475.0
1	775635.0	8409.0	45453.0
2	775132.0	8634.0	45373.0
3	772351.0	8970.0	45293.0
4	770821.0	9359.0	45214.0
..	...	...	...
95	467773.0	19687.0	31038.0
96	462535.0	19302.0	30749.0
97	457619.0	18807.0	30469.0
98	454225.0	18437.0	30274.0
99	451796.0	18249.0	30109.0

	onVentilatorCurrently ...	totalTestResults	lastModified \
0	2802.0 ...	363825123	2021-03-07T24:00:00Z
1	2811.0 ...	362655064	2021-03-06T24:00:00Z
2	2889.0 ...	361224072	2021-03-05T24:00:00Z
3	2973.0 ...	359479655	2021-03-04T24:00:00Z
4	3094.0 ...	357888671	2021-03-03T24:00:00Z
..		...	...
95	6855.0 ...	201554613	2020-12-02T24:00:00Z
96	6651.0 ...	199966644	2020-12-01T24:00:00Z
97	6520.0 ...	198472598	2020-11-30T24:00:00Z
98	6245.0 ...	196952358	2020-11-29T24:00:00Z
99	6148.0 ...	195615223	2020-11-28T24:00:00Z

	recovered	total	posNeg	deathIncrease	hospitalizedIncrease \
0	None	0	0	842	726
1	None	0	0	1680	503
2	None	0	0	2221	2781
3	None	0	0	1743	1530
4	None	0	0	2449	2172
..	...	...	...	...	...
95	None	0	0	2811	5238
96	None	0	0	2489	4916
97	None	0	0	1037	3394
98	None	0	0	825	2429
99	None	0	0	1243	3485

	negativeIncrease	positiveIncrease	totalTestResultsIncrease
0	131835	41835	1170059
1	143835	60015	1430992
2	271917	68787	1744417
3	177957	65487	1590984
4	267001	66836	1406795
..	...	...	...

95	-658774	203429	1587969
96	263699	181183	1494046
97	344231	150031	1520240
98	226788	137254	1337135
99	324413	151469	1675224

[100 rows x 24 columns]

1.0.1 Step 1. Dropping Duplicates from dataframe

```
[119]: pd.set_option('display.max_rows', 180)
(df
  .filter(['date'])
  .drop_duplicates()
)
```

```
[119]:      date
0    20210307
1    20210306
2    20210305
3    20210304
4    20210303
..      ...
415  20200117
416  20200116
417  20200115
418  20200114
419  20200113
```

[420 rows x 1 columns]

1.0.2 Step 2. Format Date and find Total number of negativeIncrease by month

```
[179]: df['date_new'] = pd.to_datetime(df['date'], format='%Y%m%d')

monthly_sum = df.groupby(pd.Grouper(key='date_new',
    ↪freq='ME'))['negativeIncrease'].sum()

print(monthly_sum)
```

```
date_new
2020-01-31      0
2020-02-29      0
2020-03-31    460602
2020-04-30    1843007
2020-05-31    4171471
2020-06-30    5772802
2020-07-31    7684224
```

```

2020-08-31    7136506
2020-09-30    6684185
2020-10-31    7522627
2020-11-30    9225202
2020-12-31    8143547
2021-01-31    8527310
2021-02-28    6044941
2021-03-31    1366401
Freq: ME, Name: negativeIncrease, dtype: int64

```

1.0.3 Step 3: Drop Rows Based on Specific Columns

```

[185]: df_clean = df.dropna(subset=['positive', 'negative', 'pending', 'hospitalizedCurrently', 'hospitalizedCumulative', 'inIcuCurrently', 'inIcuCumulative', 'onVentilatorCurrently', 'lastModified', 'recovered', 'total'])
print(df_clean)

```

	date	states	positive	negative	pending	hospitalizedCurrently	\
0	20210307	56	28756489.0	74582825.0	11808.0	40199.0	
1	20210306	56	28714654.0	74450990.0	11783.0	41401.0	
2	20210305	56	28654639.0	74307155.0	12213.0	42541.0	
3	20210304	56	28585852.0	74035238.0	12405.0	44172.0	
4	20210303	56	28520365.0	73857281.0	11778.0	45462.0	
..	...	...	...	...	...	...	
342	20200330	56	171965.0	416268.0	65369.0	15892.0	
343	20200329	56	150826.0	366103.0	65404.0	14055.0	
344	20200328	56	131143.0	324271.0	65698.0	12393.0	
345	20200327	56	111358.0	255808.0	59795.0	10887.0	
346	20200326	56	92143.0	210652.0	60131.0	7805.0	

	hospitalizedCumulative	inIcuCurrently	inIcuCumulative	\
0	776361.0	8134.0	45475.0	
1	775635.0	8409.0	45453.0	
2	775132.0	8634.0	45373.0	
3	772351.0	8970.0	45293.0	
4	770821.0	9359.0	45214.0	
..	...	...	...	
342	5208.0	3087.0	187.0	
343	4624.0	2456.0	156.0	
344	4065.0	2174.0	140.0	
345	3383.0	1792.0	124.0	
346	2606.0	1299.0	91.0	

	onVentilatorCurrently	...	lastModified	recovered	total	\
0	2802.0	...	2021-03-07T24:00:00Z	None	0	
1	2811.0	...	2021-03-06T24:00:00Z	None	0	
2	2889.0	...	2021-03-05T24:00:00Z	None	0	
3	2973.0	...	2021-03-04T24:00:00Z	None	0	
4	3094.0	...	2021-03-03T24:00:00Z	None	0	

```

..          ... ..
342          449.0 ... 2020-03-30T24:00:00Z      None      0
343          433.0 ... 2020-03-29T24:00:00Z      None      0
344          390.0 ... 2020-03-28T24:00:00Z      None      0
345          293.0 ... 2020-03-27T24:00:00Z      None      0
346          258.0 ... 2020-03-26T24:00:00Z      None      0

```

```

      posNeg  deathIncrease hospitalizedIncrease negativeIncrease \
0           0           842           726           131835
1           0          1680           503           143835
2           0          2221          2781           271917
3           0          1743          1530           177957
4           0          2449          2172           267001
..          ...          ...          ...          ...
342         0           584           584           50165
343         0           506           559           41832
344         0           550           682           68463
345         0           411           777           45156
346         0           313           655           42359

```

```

      positiveIncrease  totalTestResultsIncrease  date_new
0           41835           1170059 2021-03-07
1           60015           1430992 2021-03-06
2           68787           1744417 2021-03-05
3           65487           1590984 2021-03-04
4           66836           1406795 2021-03-03
..          ...          ...          ...
342          21139           100592 2020-03-30
343          19683           86974 2020-03-29
344          19785           111430 2020-03-28
345          19215           101260 2020-03-27
346          17592           106945 2020-03-26

```

[347 rows x 25 columns]

1.0.4 Step 4: Drop non required Columns

```

[194]: df_clean.drop(columns=['recovered', 'total', 'posNeg'], inplace=True)
print(df_clean)

```

```

      date  states  positive  negative  pending  hospitalizedCurrently \
0  20210307     56  28756489.0  74582825.0  11808.0           40199.0
1  20210306     56  28714654.0  74450990.0  11783.0           41401.0
2  20210305     56  28654639.0  74307155.0  12213.0           42541.0
3  20210304     56  28585852.0  74035238.0  12405.0           44172.0
4  20210303     56  28520365.0  73857281.0  11778.0           45462.0
..          ...    ...          ...          ...          ...
342 20200330     56   171965.0   416268.0  65369.0           15892.0

```

343	20200329	56	150826.0	366103.0	65404.0	14055.0
344	20200328	56	131143.0	324271.0	65698.0	12393.0
345	20200327	56	111358.0	255808.0	59795.0	10887.0
346	20200326	56	92143.0	210652.0	60131.0	7805.0

	hospitalizedCumulative	inIcuCurrently	inIcuCumulative	\
0	776361.0	8134.0	45475.0	
1	775635.0	8409.0	45453.0	
2	775132.0	8634.0	45373.0	
3	772351.0	8970.0	45293.0	
4	770821.0	9359.0	45214.0	
..	...	...	...	
342	5208.0	3087.0	187.0	
343	4624.0	2456.0	156.0	
344	4065.0	2174.0	140.0	
345	3383.0	1792.0	124.0	
346	2606.0	1299.0	91.0	

	onVentilatorCurrently	...	death	hospitalized	totalTestResults	\
0	2802.0	...	515151.0	776361.0	363825123	
1	2811.0	...	514309.0	775635.0	362655064	
2	2889.0	...	512629.0	775132.0	361224072	
3	2973.0	...	510408.0	772351.0	359479655	
4	3094.0	...	508665.0	770821.0	357888671	
..	...	...	...	...	...	
342	449.0	...	3422.0	5208.0	1068981	
343	433.0	...	2838.0	4624.0	968389	
344	390.0	...	2332.0	4065.0	881415	
345	293.0	...	1782.0	3383.0	769985	
346	258.0	...	1371.0	2606.0	668725	

	lastModified	deathIncrease	hospitalizedIncrease	\
0	2021-03-07T24:00:00Z	842	726	
1	2021-03-06T24:00:00Z	1680	503	
2	2021-03-05T24:00:00Z	2221	2781	
3	2021-03-04T24:00:00Z	1743	1530	
4	2021-03-03T24:00:00Z	2449	2172	
..	...	...	...	
342	2020-03-30T24:00:00Z	584	584	
343	2020-03-29T24:00:00Z	506	559	
344	2020-03-28T24:00:00Z	550	682	
345	2020-03-27T24:00:00Z	411	777	
346	2020-03-26T24:00:00Z	313	655	

	negativeIncrease	positiveIncrease	totalTestResultsIncrease	date_new
0	131835	41835	1170059	2021-03-07
1	143835	60015	1430992	2021-03-06
2	271917	68787	1744417	2021-03-05

3	177957	65487	1590984	2021-03-04
4	267001	66836	1406795	2021-03-03
..	...	...	...	...
342	50165	21139	100592	2020-03-30
343	41832	19683	86974	2020-03-29
344	68463	19785	111430	2020-03-28
345	45156	19215	101260	2020-03-27
346	42359	17592	106945	2020-03-26

[347 rows x 22 columns]

/var/folders/18/ygzmjvj90m72g8_1d1sfk1b80000gn/T/ipykernel_24787/1582743883.py:1

: SettingWithCopyWarning:

A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```
df_clean.drop(columns=['recovered', 'total', 'posNeg'], inplace=True)
```

1.0.5 Step 5: sort data by new date column

```
[202]: df_clean = df_clean.sort_values(by='date_new', ascending=False)
df_clean.head(10)
```

```
[202]:
```

	date	states	positive	negative	pending	hospitalizedCurrently	\
0	20210307	56	28756489.0	74582825.0	11808.0	40199.0	
1	20210306	56	28714654.0	74450990.0	11783.0	41401.0	
2	20210305	56	28654639.0	74307155.0	12213.0	42541.0	
3	20210304	56	28585852.0	74035238.0	12405.0	44172.0	
4	20210303	56	28520365.0	73857281.0	11778.0	45462.0	
5	20210302	56	28453529.0	73590280.0	11196.0	46388.0	
6	20210301	56	28399281.0	73334501.0	11748.0	46738.0	
7	20210228	56	28351189.0	73216424.0	11708.0	47352.0	
8	20210227	56	28296840.0	73012825.0	11731.0	48871.0	
9	20210226	56	28225595.0	72807735.0	11945.0	51112.0	

	hospitalizedCumulative	inIcuCurrently	inIcuCumulative	\
0	776361.0	8134.0	45475.0	
1	775635.0	8409.0	45453.0	
2	775132.0	8634.0	45373.0	
3	772351.0	8970.0	45293.0	
4	770821.0	9359.0	45214.0	
5	768649.0	9465.0	45084.0	
6	766778.0	9595.0	44956.0	
7	765754.0	9802.0	44907.0	
8	764875.0	10114.0	44875.0	
9	763447.0	10466.0	44791.0	

	onVentilatorCurrently	...	death	hospitalized	totalTestResults	\
0	2802.0	...	515151.0	776361.0	363825123	
1	2811.0	...	514309.0	775635.0	362655064	
2	2889.0	...	512629.0	775132.0	361224072	
3	2973.0	...	510408.0	772351.0	359479655	
4	3094.0	...	508665.0	770821.0	357888671	
5	3169.0	...	506216.0	768649.0	356481876	
6	3171.0	...	504488.0	766778.0	355138357	
7	3245.0	...	503247.0	765754.0	353983917	
8	3335.0	...	502196.0	764875.0	352575495	
9	3466.0	...	500349.0	763447.0	350920316	

	lastModified	deathIncrease	hospitalizedIncrease	negativeIncrease	\
0	2021-03-07T24:00:00Z	842	726	131835	
1	2021-03-06T24:00:00Z	1680	503	143835	
2	2021-03-05T24:00:00Z	2221	2781	271917	
3	2021-03-04T24:00:00Z	1743	1530	177957	
4	2021-03-03T24:00:00Z	2449	2172	267001	
5	2021-03-02T24:00:00Z	1728	1871	255779	
6	2021-03-01T24:00:00Z	1241	1024	118077	
7	2021-02-28T24:00:00Z	1051	879	203599	
8	2021-02-27T24:00:00Z	1847	1428	205090	
9	2021-02-26T24:00:00Z	2141	1868	276829	

	positiveIncrease	totalTestResultsIncrease	date_new
0	41835	1170059	2021-03-07
1	60015	1430992	2021-03-06
2	68787	1744417	2021-03-05
3	65487	1590984	2021-03-04
4	66836	1406795	2021-03-03
5	54248	1343519	2021-03-02
6	48092	1154440	2021-03-01
7	54349	1408422	2021-02-28
8	71245	1655179	2021-02-27
9	74857	1803309	2021-02-26

[10 rows x 22 columns]

[]: